

FirstKnock GitHub & Open Source Tools Audit

FirstKnock Sales OS

GitHub & Open Source Tools Audit — April 2026

April 2026

CONFIDENTIAL — INTERNAL USE ONLY

4Layers Audited	15Repos Evaluated	3Templates Assessed	8Sections Covered
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SECTION 1: EXECUTIVE BRIEF — WHAT THIS AUDIT COVERS

FirstKnock is not building a side project. It is building a full-stack, field-intelligence Sales OS for door-to-door sales reps — a machine that optimizes walking routes, automates cold outreach, generates viral training content, and scales LinkedIn prospecting without burning the founder's account. This audit exists because the open-source ecosystem is full of vaporware, abandoned repos, and tools that look great in a README but collapse under production load. We are not here to collect GitHub stars. We are here to ship.

This audit evaluates 15 GitHub repositories and 3 n8n workflow templates across four critical layers of the FirstKnock Sales OS stack. Every tool was assessed against the same criteria: GitHub maturity (stars, forks, commit velocity, contributor count), technical capability depth, integration ease for a 2–3 person engineering team, license risk, and direct fit for FirstKnock's specific use cases — optimized walking routes for D2D reps visiting recently sold homes, automated B2C and B2B email outreach, short-form video content for TikTok and Reels, and LinkedIn prospecting for pest control and smart home security dealers.

The methodology was direct: pull verified GitHub stats, cross-reference documentation against real deployment requirements, and apply a ruthless fit-scoring rubric. Tools that are closed-source, commercially upselling, or simply not production-ready received explicit DO NOT USE verdicts. No tool gets a pass because it has a slick landing page.

Audit Scope Summary

- Layer 1 — Route Optimization: VROOM, Valhalla, Nextroute
- Evaluating VRP solver + routing engine combinations for 10–50 stop walking routes in residential neighborhoods
- Layer 2 — Cold Email Automation: ColdFlow, CybrOutreach, cold-cli, OutreachStudio
- Evaluating AI personalization, sequence management, deliverability infrastructure, and sending engines for B2C homeowner nurture and B2B dealer outreach

- Layer 3 — Video Generation: AI-Automated-Short-Video-Generator, TikTokAIVideoGenerator, short-video-maker, tiktok-agent
- Evaluating automated pipelines for D2D training content, testimonials, and brand reels
- Layer 4 — LinkedIn & Social Outreach: OpenOutreach + broader automation risk landscape
- Evaluating automation safety, detection risk, and founder account protection protocols in a post-2026 LinkedIn enforcement environment
- Layer 5 — n8n Workflow Automation: Platform assessment + 3 workflow templates
- Evaluating n8n as the connective tissue binding all four layers together

Key Findings at a Glance

- SHIP IT: Valhalla + VROOM is the non-negotiable route optimization stack. Both are MIT/open-source, Docker-deployable, and natively integrate with each other.
- SHIP IT: CybrOutreach (content generation) + Mautic (automation) + Postal (infrastructure) is the composite cold email stack. No single tool does it all.
- SHIP IT: TikTokAIVideoGenerator is the most production-ready video generation repo. Pair it with the n8n Shorts Generator template for a fully automated content pipeline.
- AVOID: OutreachStudio is BSL-licensed vaporware with 9 total commits and no public code. Do not build on it.
- CAUTION: LinkedIn automation carries a 23% account restriction rate within 90 days as of 2026. OpenOutreach is technically capable but not safe for founder accounts.
- FOUNDATION: n8n (self-hosted) is the automation backbone. 160K+ GitHub stars, 230K+ active users, unlimited executions for a flat \$5–50/month VPS cost.

This document is designed to be handed directly to a developer with the instruction: build this. Every recommendation includes a specific rationale, integration architecture, and risk mitigation strategy. Vague suggestions have been eliminated.

SECTION 2: LAYER 1 — ROUTE OPTIMIZATION: BUILD THE FASTEST ROUTES ON THE BLOCK

FirstKnock's core value proposition lives or dies on route quality. A sales rep who walks an inefficient route wastes 30–45 minutes per day — time that compounds into lost revenue across a team of 10+ reps. The route optimization layer must solve a Vehicle Routing Problem (VRP) with pedestrian-specific travel times, support job priorities (most recently sold homes first), and return results in under 2 seconds for 10–50 stops. Here is what the open-source ecosystem actually delivers.

Repo 1: VROOM (vroom-project/vroom)

Stars / Forks	1,700 stars / 405 forks
Language	C++ (99%) — C++20 standard
Contributors / Open Issues	36+ contributors / 37 open issues
Last Commit / Latest Release	March 17, 2026 / v1.15.0 (March 12, 2026)
License	Open Source (maintained by Verso)

GitHub Maturity Assessment

VROOM is a mature, actively maintained project. A release as recent as March 12, 2026 (v1.15.0) with a commit just 5 days later signals a team that ships continuously. 36+ contributors and 405 forks indicate a healthy ecosystem of downstream users and integrators. 37 open issues is a healthy signal — not zero (which would mean no one is using it), and not 500 (which would mean it's drowning). This is a production-grade codebase.

Technical Capabilities

- Solves TSP,CVRP,VRPTW, MDHVRPTW, and PDPTW problem types — covers every routing scenario FirstKnock will encounter
- C++20 implementation with counting_semaphore for multithreaded parallel routing — defaults to 128 parallel requests
- SwapStar speedup and local search optimizations deliver near-instant results for small route sets (10–50 stops)
- Native integration with Valhalla, OSRM, and ORS routing engines via wrapper — no custom glue code required
- REST API via vroom-express wrapper — JSON in, optimized route JSON out
- Docker-deployable — single docker-compose.yml gets you running in under 10 minutes
- Supports job priorities, time windows, vehicle-specific costs, and per-vehicle task times
- Heuristic regrets for improved solution accuracy on partial solutions

Table 1 — VROOM Assessment Scorecard

Dimension	Score	Justification
Integration Ease	9/10	REST API via vroom-express, Docker-native, native Valhalla wrapper. Minimal custom code required.
FirstKnock Fit	9/10	Solves VRP with priorities and time windows. Handles 10–50 stops in milliseconds. Perfect for daily rep route generation.
Maturity	9/10	v1.15.0 released March 2026, 36+ contributors, 1,700 stars, active CI/CD pipeline.
License Safety	10/10	Fully open source. No commercial upsell. No usage restrictions.

Recommended Use Case in FirstKnock Stack

VROOM is the optimization brain. Every morning, when a rep's daily stop list is finalized (recently sold homes within their territory), the FirstKnock backend sends a JSON payload to the vroom-express REST API containing the list of stops with coordinates, priorities, and time windows. VROOM calls Valhalla for the pedestrian walking matrix, runs the VRP solver, and returns the optimally ordered stop sequence in under 500ms. The ordered stops are then pushed to the mobile app.

Key Risks & Limitations

- VROOM is a solver only — it has zero routing intelligence without an external engine (Valhalla). Do not deploy VROOM without Valhalla.
- C++20 build complexity: if you need to compile from source (vs. Docker), you need a developer comfortable with CMake and C++ toolchains.
- vroom-express is a Node.js wrapper — adds a JavaScript dependency to an otherwise C++ stack. Monitor for version drift.

Repo 2: Valhalla (valhalla/valhalla)

Stars / Forks	5,600 stars / 878 forks
Language	C++ (94.6%)
Contributors / Open Issues	254 contributors / 769 open issues
Last Commit / Latest Release	April 3, 2026 / v3.6.3 (February 19, 2026)
License / Total Commits	MIT / 14,512 commits

GitHub Maturity Assessment

Valhalla is the most mature project in this entire audit. 14,512 total commits, 254 contributors, 5,600 stars, and a commit as recent as April 3, 2026 — this is enterprise-grade infrastructure with a massive community behind it. The 769 open issues is the only yellow flag, but for a project of this scale and complexity, that number is proportionate. Valhalla is used in production by major mapping and logistics companies worldwide. It is not going anywhere.

Technical Capabilities

- Native pedestrian routing mode — understands sidewalks, crosswalks, cul-de-sacs, and walking-specific shortcuts from OSM data
- Multi-modal support: auto, bicycle, pedestrian, transit (GTFS), winter roads, ferries
- Tile-based OSM data architecture — download only the metro areas you need, keeping storage and RAM requirements manageable
- Offline-capable — no external API calls once tiles are downloaded. Full data sovereignty.
- Mobile SDKs for iOS and Android — enables turn-by-turn walking navigation directly in the FirstKnock app
- Python bindings (pyvalhalla) and NodeJS bindings for backend integration
- Rich turn-by-turn narrative directions in multiple languages
- Docker deployment with public demo at valhalla.openstreetmap.de for testing
- Modular architecture: Loki (request handling), Thor (route calculation), Odin (narrative generation), Tyr (API interfacing)

Table 2 — Valhalla Assessment Scorecard

Dimension	Score	Justification
Integration Ease	8/10	Docker deployment is straightforward. Tile building for new metro areas requires significant processing time — plan for this in your deployment pipeline.
FirstKnock Fit	10/10	The only routing engine in this audit with native pedestrian mode and OSM residential neighborhood awareness. Non-negotiable for D2D walking routes.
Maturity	10/10	14,512 commits, 254 contributors, MIT license, v3.6.3 released February 2026. The gold standard of open-source routing.
License Safety	10/10	MIT license. Use it, modify it, ship it. No restrictions.

Recommended Use Case in FirstKnock Stack

Valhalla is the routing intelligence layer. It generates the pedestrian walking time and distance matrix that VROOM consumes. It also powers the turn-by-turn navigation experience in the FirstKnock mobile app between optimized stops. Deploy Valhalla via Docker on a private server, downloading OSM tile data only for your active target metro areas. This keeps the tile storage footprint small and RAM requirements manageable during early-stage operations.

Key Risks & Limitations

- Tile building is computationally heavy for large regions. For a full US state, expect hours of processing time. Start with city-level tiles.
- 769 open issues — monitor for any issues affecting pedestrian routing or tile-level exclusion logic (known bug affecting transit isochrones).
- C++ build complexity if compiling from source. Use the official Docker image to avoid this entirely.

Repo 3: Nextroute (nextmv-io/nextroute)

Stars / Forks	23 stars / 7 forks
Language	Go (93.8%), Python (6.2%)
Contributors / Open Issues	13 contributors / 1 open issue
Last Commit / Latest Release	February 17, 2026 / v1.12.3 (November 10, 2025)
License	Source-Available — NOT open source (see LICENSE)

VERDICT: PASS — DO NOT INTEGRATE. Nextroute's license is classified as 'Source-Available (NOASSERTION)' — it is NOT open source. The Nextmv platform requires commercial cloud access for map data, creating a vendor lock-in risk that is unacceptable for a lean startup. With only 23 stars and a latest release from November 2025, the community footprint does not justify the licensing risk.

Nextroute does offer impressive constraint modeling — 30+ constraint types including time windows, capacities, precedence, cluster constraints, and stop groups. The Python bindings are a genuine advantage for a Python-heavy team. But none of that matters when the license prohibits commercial use without a Nextmv Cloud subscription, and the map data layer requires their commercial platform. For FirstKnock's 10–50 stop use case, VROOM solves the same problem faster, cheaper, and with zero license risk.

Recommended Stack: Valhalla + VROOM Architecture

The optimal route optimization architecture for FirstKnock combines Valhalla as the routing intelligence layer and VROOM as the optimization solver. This is not a theoretical recommendation — VROOM natively integrates with Valhalla via its routing engine wrapper, meaning the two tools are designed to work together out of the box.

Integration Data Flow

1. FirstKnock Backend receives rep's daily stop list (10–50 recently sold home coordinates + priority scores)
2. Backend sends JSON payload to vroom-express REST API (POST /)
3. Payload includes: job list with coordinates, priorities, time windows, and vehicle profile (pedestrian)

4. VROOM calls Valhalla routing engine to compute the pedestrian walking time/distance matrix for all stop pairs
5. Valhalla queries its locally hosted OSM tile data — no external API calls, full offline capability
6. VROOM runs VRP solver against the matrix — returns optimally ordered stop sequence in <500ms for 50 stops
7. Ordered stop list is returned to FirstKnock backend via JSON response
8. Backend pushes ordered stops to mobile app
9. Mobile app uses Valhalla's iOS/Android SDK for turn-by-turn pedestrian navigation between stops

Deployment Architecture

- Single VPS (4 vCPU, 8GB RAM recommended) running Docker Compose
- Container 1: Valhalla — serves routing matrix requests on internal port 8002
- Container 2: vroom-express — serves optimization requests on port 3000, configured to call Valhalla internally
- Container 3: FirstKnock API backend — calls vroom-express, receives results, pushes to mobile
- OSM tile data stored in a Docker volume — download city-level tiles only for active markets
- Estimated monthly infrastructure cost: \$20–40/month on Hetzner or DigitalOcean

SECTION 3: LAYER 2 — COLD EMAIL AUTOMATION: FLOOD THE INBOX OR GO HOME

The cold email layer is where the open-source ecosystem shows its cracks. There is no single open-source tool that delivers AI personalization, domain rotation, sequence automation, and high-volume sending in one package. The four repos evaluated here range from a promising TypeScript project with a questionable availability status to outright vaporware with a Business Source License. The honest answer is that FirstKnock needs a composite stack — and this section tells you exactly how to build it.

Repo 1: ColdFlow (pypes-dev/coldflow)

Stars / Forks	7 stars / 1 fork
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Language	TypeScript (94.2%)
Contributors / Open Issues	3 contributors / 0 open issues
Last Commit / Total Commits	December 13, 2025 / 41 commits
License	GPL-3.0

GitHub Maturity Assessment

ColdFlow has the right feature vision — domain setup, list creation, lead enrichment, sequencing, campaign metrics, SPF/DKIM/DMARC checks, and a Dockerfile for containerized deployment. On paper, it is the closest thing to a self-contained cold email platform in this audit. In practice, the repository may have been made private or deleted — recent checks indicate it is inaccessible. 7 stars, 41 commits, and a last commit of December 13, 2025 (4+ months ago) paint a picture of a project that stalled before reaching production readiness.

Technical Capabilities

- Domain setup and deliverability configuration (SPF, DKIM, DMARC checks built-in)
- Lead list creation and basic enrichment pipeline
- Email sequencing with campaign metrics tracking
- Dockerfile for containerized deployment
- Google SMTP integration for sending
- AI personalization claimed but not documented — no model, prompt, or implementation details available

VERDICT: DO NOT BUILD ON COLDFLOW. The repository is potentially inaccessible (private or deleted). GPL-3.0 license means any derivative work must also be GPL-3.0 — a serious risk for a commercial product. Even if the repo becomes accessible, 41 commits and 7 stars is not a foundation you want under a production email system.

Table 3 — ColdFlow Assessment Scorecard

Dimension	Score	Justification
Integration Ease	3/10	Potentially inaccessible repo. GPL-3.0 license creates legal risk for commercial use.
FirstKnock Fit	2/10	Feature set is promising but the project is stalled, potentially private, and GPL-licensed.
Maturity	2/10	41 commits, 7 stars, last commit December 2025. Not production-ready.
License Safety	2/10	GPL-3.0 — copyleft license requires derivative works to be GPL. Unacceptable for a commercial SaaS.

Repo 2: CybrOutreach (M4ST3R-C0NTR0L/CybrOutreach)

Stars / Forks	0 stars / 0 forks
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Language	Python (100%)
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Contributors / Open Issues	1 contributor / 0 open issues
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Last Commit / Total Commits	March 10, 2026 / 5 commits
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License	MIT
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GitHub Maturity Assessment

CybrOutreach is brand new — initial v1.0 released February 17, 2026, rebranded from 'IceBreaker' on March 10, 2026. Zero stars, zero forks, 5 total commits, and a single contributor. By every traditional maturity metric, this is a pre-alpha project. But here is why it makes the recommended stack anyway: it does one thing extremely well, it is MIT-licensed, it is pure Python, and it is designed to be used as a microservice. The feature set is precisely what FirstKnock needs for

Technical Capabilities

- AIDA (Attention-Interest-Desire-Action), PAS (Problem-Agitate-Solve), and BAB (Before-After-Bridge) copywriting frameworks
- 4 tone modes: Professional, Casual, Aggressive, Friendly — selectable per campaign
- Batch CSV processing — ingest a list of leads, output personalized emails at scale
- Multi-provider AI support: OpenRouter, OpenAI, and Anthropic APIs — swap models without rewriting code
- CLI built with Click + Rich — clean developer experience, easy to wrap in a FastAPI microservice
- CONTENT GENERATOR ONLY — no sending infrastructure, no sequence management, no deliverability features

Table 4 — CybrOutreach Assessment Scorecard

Dimension	Score	Justification
Integration Ease	8/10	Pure Python, MIT license, CLI-based. Wrap in FastAPI and call it from n8n in under a day.
FirstKnock Fit	7/10	Excellent for B2B pest control/security dealer outreach. PAS framework is ideal for value-prop cold emails. Needs a sending layer.
Maturity	3/10	5 commits, 0 stars, single contributor. Treat as a utility script, not a platform dependency.
License Safety	10/10	MIT license. No restrictions. Fork it, modify it, ship it.

Repo 3: cold-cli (andersmyrmel/cold-cli)

Stars / Forks	0 stars / 0 forks
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Language	Go (100%)
Contributors / Open Issues	2 contributors (andersmyrmel + Claude) / 0 open issues
Last Commit / Total Commits	April 3, 2026 / 61 commits
License	MIT

GitHub Maturity Assessment

cold-cli is the most actively developed of the four cold email repos — 61 commits with a last commit of April 3, 2026 (the most recent in this entire audit). The 'Agent-first' design philosophy (built to work with Claude Code, Cursor, and similar AI coding agents) is a genuine differentiator. The single Go binary with SQLite storage is an elegant, zero-dependency deployment model. However, the Gmail-only constraint is a hard blocker for production cold outreach at scale.

Technical Capabilities

- Single Go binary — zero runtime dependencies, deploys anywhere
- SQLite storage — no database server required
- Gmail API integration via gws — multi-account round-robin sending
- Reply and bounce detection — automatically removes responders from sequences
- A/B testing for subject lines and body copy
- Domain deliverability diagnostics — SPF, DKIM, DMARC validation
- Unsubscribe detection and campaign cloning
- Agent-first design — works natively with Claude Code and Cursor for AI-assisted campaign management
- GMAIL ONLY — no multi-provider SMTP, no Mailgun, no SendGrid, no Postal integration

VERDICT: USEFUL FOR DEVELOPER TESTING, NOT PRODUCTION SCALE. cold-cli is an excellent tool for a developer to run small-scale outreach campaigns from their own Gmail accounts during early validation. The Gmail-only constraint and CLI-based workflow make it unsuitable as the primary sending engine for a sales team. Use it for founder-led outreach in the first 30 days, then migrate to Mautic + Postal.

Table 5 — cold-cli Assessment Scorecard

Dimension	Score	Justification
Integration Ease	6/10	Single binary is easy to deploy. Gmail-only constraint limits production scalability. No REST API — CLI only.
FirstKnock Fit	5/10	Good for founder-led early outreach. Not suitable as the team's primary sending infrastructure.
Maturity	6/10	61 commits, April 2026 last commit. Most actively developed of the four, but zero community adoption.
License Safety	10/10	MIT license. No restrictions.

Repo 4: OutreachStudio (OutreachStud-io/studio)

Stars / Forks	38 stars / 0 forks
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Language	N/A — code not public
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Contributors / Open Issues	1 contributor / 0 open issues
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Last Commit / Total Commits	October 7, 2025 / 9 commits
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License	Business Source License (BSL) — NOT open source
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VERDICT: DO NOT USE — VAPORWARE. OutreachStudio has 9 total commits, 0 forks, code that is not publicly available, a Business Source License (which is explicitly NOT open source), and a last commit from October 2025. The 38 stars are from people who starred a README. There is nothing here to integrate. Move on.

Recommended Composite Stack: Mautic + Postal + CybrOutreach

Because no single tool in this audit delivers the full cold email stack, FirstKnock must assemble a

composite architecture. The recommended stack uses three proven open-source tools in distinct roles: Postal as the deliverability infrastructure, Mautic as the automation and sequence engine, and CybrOutreach as the AI personalization microservice.

Use Case 1: B2C Homeowner Follow-Up (Post-Door-Knock Nurture)

When a rep completes a door knock and logs the interaction in the FirstKnock mobile app, a webhook fires to Mautic, entering the homeowner into a localized nurture sequence. The sequence delivers 3–5 emails over 7–14 days: a same-day 'nice to meet you' email, a neighborhood social proof email (recent sales in their area), a value-add email (home security tips, pest prevention checklist), and a soft CTA. Mautic handles the trigger logic, sequence timing, and open/click tracking. Postal handles the actual SMTP delivery, managing IP reputation and bounce handling.

Use Case 2: B2B Pest Control / Security Dealer Outreach

For B2B outreach to pest control owners and smart home security dealers, CybrOutreach ingests a CSV of target businesses, uses the PAS (Problem-Agitate-Solve) framework with Anthropic Claude or OpenAI GPT-4o to generate highly personalized cold emails, and outputs a new CSV with finalized copy. That CSV is imported into Mautic via API as a campaign, with the AI-generated opener stored as a custom field. Mautic sends the emails through Postal with proper SMTP routing.

Deliverability Strategy

- Domain Separation: Use a dedicated sending subdomain (e.g., mail.firstknock.com) — never send cold outreach from your primary domain
- SPF, DKIM, DMARC: Configure all three on your sending domain before sending a single email. Postal provides tooling for this.
- Domain Warming: Start at 20–30 emails/day per domain, doubling weekly over 4 weeks before reaching full volume
- Sending Limits: Cap at 200 emails/day per domain during warm-up. Scale to 500/day after 30 days of positive engagement signals.
- Bounce Management: Postal's built-in bounce handling automatically suppresses hard bounces. Target <2% bounce rate.
- Engagement Signals: Monitor open rates (target >25%) and reply rates (target >3%) as primary deliverability health metrics

SECTION 4: LAYER 3 — VIDEO GENERATION: GO VIRAL OR GO HOME

FirstKnock's content strategy requires a repeatable pipeline for generating D2D sales training videos, rep highlight reels, and brand content for TikTok and Instagram Reels. The four repos evaluated here represent the current state of open-source AI video generation — a space that is moving fast but still requires significant assembly. Here is what actually works.

Stars / Forks	12 stars / 3 forks
Language	Python (100%)
Contributors	1 contributor
Last Commit / Total Commits	February 12, 2026 / 11 commits
License	Not specified

Technical Capabilities

- Full pipeline: script generation → TTS → video editing → TikTok upload
- Llama3 via Groq API for script generation — fast, cheap inference
- Coqui TTS with Edge TTS fallback for voiceovers
- Reddit fact scraping for content sourcing — not relevant for FirstKnock's use case
- MoviePy for video editing and assembly
- TikTok API auto-upload and Mega.nz backup
- No license specified — legal risk for commercial use

Table 6 — AI-Automated-Short-Video-Generator Assessment Scorecard

Dimension	Score	Justification
Integration Ease	6/10	Pure Python pipeline. Reddit scraping is irrelevant and must be replaced with FirstKnock content sources.
FirstKnock Fit	6/10	Good pipeline architecture but Reddit-focused content sourcing needs replacement. No license is a red flag.
Maturity	4/10	11 commits, 12 stars, single contributor, no license. Treat as a reference implementation.

License Safety	3/10	No license specified means all rights reserved by default. Cannot legally use in commercial product without permission.
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Repo 2: TikTokAIVideoGenerator (GabrielLaxy)

Stars / Forks	54 stars / 15 forks
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Language	Python (99.7%)
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Contributors	1 contributor
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Last Commit / Total Commits	March 4, 2025 / 15 commits
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License	CC0-1.0 (Public Domain)
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Technical Capabilities

- Llama3 via Groq API for script generation
- FLUX-1 for AI image generation — high-quality visual assets
- Kokoro TTS with Edge TTS fallback — natural-sounding voiceovers
- Whisper for automatic caption generation
- MoviePy for video assembly and editing
- Output: 1080x1920 (vertical), 24fps, H.264/AAC encoding, max 5 words/line subtitles — production-ready format
- CC0-1.0 (Public Domain) license — zero restrictions, use commercially without attribution

Table 7 — TikTokAIVideoGenerator Assessment Scorecard

Dimension	Score	Justification
Integration Ease	7/10	Well-structured Python pipeline. FLUX-1 image generation adds API cost but produces superior visuals.

FirstKnock Fit	8/10	Most starred of the four. Production-ready output format. CC0 license is ideal. Last commit March 2025 is the only
Maturity	6/10	54 stars, 15 forks, but last commit was March 2025 — over a year ago. May need dependency updates.
License Safety	10/10	CC0-1.0 Public Domain. No restrictions whatsoever. The best possible license for commercial use.

Repo 3: short-video-maker (leke-adewa)

Stars / Forks	24 stars / 5 forks
Language	Python (100%)
Contributors	2 contributors
Last Commit / Total Commits	April 6, 2026 / 6 commits
License	MIT

Technical Capabilities

- Gemini API and ChatGPT for AI-driven script and content creation
- Remotion for programmatic video composition — React-based, highly customizable
- FFmpeg for video processing and encoding
- Platform optimization features for vertical video formats
- MIT license — clean for commercial use
- MOST RECENTLY ACTIVE of the four — last commit April 6, 2026 (today)

Table 8 — short-video-maker Assessment Scorecard

Dimension	Score	Justification
Integration Ease	7/10	Remotion adds a React/Node.js dependency but enables highly branded, template-driven video output.
FirstKnock Fit	7/10	Most recently active repo. Remotion's template system is ideal for consistent FirstKnock branding across all videos.
Maturity	5/10	6 commits total — extremely early stage. But April 2026 activity is a positive signal.
License Safety	10/10	MIT license. No restrictions for commercial use.

Repo 4: tiktok-agent (mohammedelahmar)

Stars / Forks	13 stars / 4 forks
Language	Python (74.4%), JavaScript (25.2%)
Contributors	1 contributor
Last Commit / Total Commits	February 2026 / 34 commits
License	MIT + Apache 2.0 + BSD 3-Clause

tiktok-agent serves a fundamentally different use case from the other three repos. It is not a video generation tool — it is a clip extraction and repurposing tool. Its engagement modeling for viral clip extraction, face detection, vertical video formatting, watermarking, and web UI make it ideal for taking existing long-form content (sales training webinars, rep testimonials, product demos) and automatically extracting the most engaging 30–60 second clips for TikTok and Reels.

Table 9 — tiktok-agent Assessment Scorecard

Dimension	Score	Justification
Integration Ease	7/10	Web UI lowers the barrier for non-technical team members. Python + JavaScript stack is manageable.
FirstKnock Fit	7/10	Different use case — clip repurposing, not generation. Excellent for turning sales training recordings into short-form content.
Maturity	6/10	34 commits, February 2026 activity. Most mature of the four by commit count.
License Safety	9/10	MIT + Apache 2.0 + BSD 3-Clause — all permissive licenses. No commercial restrictions.

n8n AI-Automated Shorts Generator Template Assessment

The n8n AI-Automated TikTok/YouTubeShorts/Reels Generator template is the most production-ready video automation solution in this entire audit. It earns a FirstKnock Fit Score of 9/10. The workflow takes ideas from a Google Sheet and automatically produces branded, published short-form videos using OpenAI for scripting, PiAPI + ElevenLabs for visuals and voiceovers, Creatomate for final video assembly, and Google Drive + Discord for storage and notifications.

- Google Sheets node: Sales team inputs daily content ideas (D2D tips, objection handling scripts, neighborhood stats)
- OpenAI node: Expands ideas into full scripts with hooks, body, and CTAs optimized for short-form retention
- PiAPI node: Generates visual assets (B-roll style images/clips) from script prompts
- ElevenLabs node: Converts script to natural-sounding voiceover using a consistent FirstKnock brand voice
- Creatomate node: Assembles all assets into a branded 1080x1920 video with captions, logo, and CTA overlay
- Google Drive node: Archives the final video file
- Discord node: Notifies the content team that a new video is ready for review and publishing

Recommended Content Pipeline for FirstKnock

1. **GENERATE** (n8n + TikTokAIVideoGenerator): Use the n8n Shorts Generator template for fully automated faceless training content — daily D2D tips, objection handling scripts, neighborhood market stats
2. **REPURPOSE** (tiktok-agent): Record weekly sales training sessions and team calls. Run recordings through tiktok-agent to automatically extract the best 30–60 second clips for Reels.
3. **DISTRIBUTE**: Publish generated and repurposed content to TikTok, Instagram Reels, and YouTube Shorts on a 5-day/week cadence
4. **SKIP**: Do not build a custom video generation pipeline from scratch. The n8n template + TikTokAIVideoGenerator covers 90% of FirstKnock's content needs with minimal engineering investment.

SECTION 5: LAYER 4 — LINKEDIN & SOCIAL OUTREACH: CONNECT OR GET LEFT BEHIND

LinkedIn is FirstKnock's primary channel for reaching pest control owners, smart home security dealers, and D2D sales managers. It is also the channel where a single wrong move can permanently destroy the founder's most valuable professional network. The 2026 LinkedIn enforcement environment is not the same as 2022. The rules have changed, the detection algorithms have evolved, and the consequences are severe. This section tells you exactly what the risk landscape looks like and how to navigate it without losing your account.

OpenOutreach (eracle/OpenOutreach) — Full Technical Assessment

Stars	~1,400
Language / Framework	Python / FastAPI
Deployment	Docker + Docker Compose

Browser Automation	Patchright (stealth, replaces Playwright)
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Technical Capabilities

- Profile visits, connection requests, direct messaging, InMail, post reactions, and comments
- Profile enrichment and scheduling for recurring automation tasks
- FastAPI REST interface — can be called programmatically from n8n or the FirstKnock backend
- Patchright browser automation (stealth fork of Playwright) — designed to evade LinkedIn's browser fingerprinting
- Android app-based methods available for potentially higher action limits
- Docker Compose deployment — self-hosted, no SaaS dependency
- ~1,400 GitHub stars — the most community-validated open-source LinkedIn automation tool available

The 2026 LinkedIn Enforcement Reality

CRITICAL WARNING: LinkedIn account restrictions have increased 340% since January 2026. Approximately 23% of users face account restrictions within 90 days of using automation tools. A permanent ban on the founder's LinkedIn account is an existential risk for a B2B startup. Do not use OpenOutreach on your primary founder account.

LinkedIn's detection algorithms have evolved from simple volume-based rate limiting to sophisticated behavioral analysis. The platform now builds a 'profile activity DNA' for every account — a baseline of historical activity patterns. Any sudden deviation from that baseline triggers automated review. This means that even a technically sophisticated tool like OpenOutreach, running at conservative volumes, can trigger restrictions if the account's historical baseline is low.

What Triggers Detection

- Action velocity: Sending 50 connection requests on day 1 when your historical average is 2/day
- IP and device fingerprinting: Automation running from a server IP that has never been associated with your account
- Session density: 200 profile views in 4 hours — humans don't browse that fast
- Message uniformity: Identical or near-identical messages sent to multiple recipients in a short window
- Low acceptance rates: If <20% of your connection requests are accepted, LinkedIn flags your account as spam
- Sudden activity surges after quiet periods: Going from 0 activity to 100 actions/day overnight

Recommended Approach: Semi-Automated, Behavioral Strategy

For FirstKnock's B2B outreach to pest control owners and security dealers, the recommended

approach is a semi-automated strategy that prioritizes account safety over volume. Use a dedicated secondary LinkedIn account (not the founder's primary account) for automation testing. For the founder's account, use a budget commercial tool (such as WarmySender or a similar behavioral-safe platform) that includes built-in safety ramps and behavioral randomization.

Safety Protocol: The 4-Week Ramp

Table 10 — LinkedIn Safety Ramp Protocol

Week	Connection Requests/Day	Profile Views/Day	Messages/Day	Action
Week 1	10	20	5	Warm-up only. No bulk messaging.
Week 2	20	40	10	Begin personalized connection
Week 3	30	60	20	Add follow-up messages to accepted connections.
Week 4+	40–50 max	80 max	30 max	Full cadence. Monitor acceptance rate daily.

- **HARD LIMIT:** Never exceed 80 connection requests per day, even after full warm-up
- **KILL SWITCH:** If connection acceptance rate drops below 20%, pause all automation immediately for 72 hours
- **INFRASTRUCTURE:** Use dedicated residential proxies for each automated account — never run automation from a shared server IP
- **CONTENT STRATEGY:** Shift from mass outreach to meaningful engagement — comment on posts, share relevant content, build credibility before connecting
- **FOUNDER ACCOUNT:** Use only manual or semi-manual outreach on the founder's primary account. The risk of a permanent ban is not worth the volume gain.

SECTION 6: LAYER 5 — n8n WORKFLOW AUTOMATION: THE GLUE THAT HOLDS IT ALL TOGETHER

n8n is not a nice-to-have for FirstKnock's stack — it is the connective tissue that makes every other layer function as a unified system. Without n8n, you have five isolated tools that require custom integration code to communicate. With n8n, you have a visual workflow engine that

connects Valhalla route data to the mobile app, triggers Mautic email sequences from door knock events, feeds content ideas to the video generation pipeline, and manages LinkedIn outreach cadences — all without writing a single line of custom integration code.

n8n Platform Assessment

160K+GitHub Stars	230K+Active Users	400+Native Nodes	FreeSelf-Hosted Cost
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n8n's 160,000+ GitHub stars make it one of the most community-validated open-source projects in the automation space. The platform supports over 400 pre-built nodes including native AI integrations, CRM connections, email platforms, Slack, Stripe, and every major cloud service. Custom JavaScript and Python functions can be written directly within workflows, enabling complex business logic without external microservices. Event-driven automation via webhooks, API calls, and cron jobs covers every trigger pattern FirstKnock needs.

Self-Hosted vs. Cloud Cost Analysis

Table 11 — n8n Self-Hosted vs. Cloud Cost Comparison

Feature	Self-Hosted (Docker)	n8n Cloud Starter (\$20/mo)	n8n Cloud Pro (\$50/mo)
Base Cost	Free (Community Edition)	\$20/month	\$50/month
Infrastructure Cost	\$5–50/month (VPS)	Included	Included
Execution Limits	Unlimited	2,500/month	10,000/month
Maintenance	Manual (DevOps required)	Automatic updates	Automatic updates
Data Sovereignty	Full control	n8n servers	n8n servers
Recommended For	Month 2+ (post-validation)	Weeks 1–4 (rapid prototyping)	Weeks 1–4 (higher volume)

RECOMMENDATION: Start on n8n Cloud (\$20–50/month) for the first 4 weeks to avoid DevOps overhead during rapid prototyping. Once workflows exceed 10,000 executions/month or the team has DevOps capacity, migrate to self-hosted Docker on a \$20–40/month VPS (Hetzner or DigitalOcean) for unlimited executions and full data sovereignty.

Workflow Template Assessments

Table 12 — n8n Workflow Template Assessment

Template	Nodes Involved	Complexity	FirstKnock Fit	Priority
AI-Automated TikTok/YouTube Shorts/Reels Generator	Google Sheets → OpenAI → PiAPI + ElevenLabs → Creatomate	Advanced	9/10	High — deploy in Phase 2
B2B Lead Generation + Email Campaigns (n8n.io/workflows/8269)	→ Google Lead sourcing → Drive + Discord → enrichment → Mautic API →	Intermediate	8/10	High — deploy in Phase 1
Cold Outreach with n8n + AI (dev.to/jbhflow)	AI personalization → sequence management → multi-channel delivery	Intermediate	7/10	Medium — deploy in Phase 2

Recommended n8n Architecture for FirstKnock

1. Workflow 1 — Door Knock Trigger (Deploy Week 1): Mobile app fires webhook to n8n when rep logs a door knock. n8n enriches the homeowner record, adds them to Mautic, and triggers the B2C nurture sequence via Postal.
2. Workflow 2 — Daily Route Generation (Deploy Week 1): Cron job at 6:00 AM daily. n8n pulls the rep's stop list from the FirstKnock database, sends it to vroom-express, receives the optimized route, and pushes it to the mobile app.
3. Fallback: If vroom-express is unavailable, n8n sends an alert to the team Discord and falls back to a simple distance-sorted list.
4. Workflow 3 — B2B Lead Enrichment + Outreach (Deploy Week 2): n8n pulls new B2B leads from the CRM, sends them to the CybrOutreach Python microservice for AI email generation, imports the personalized copy into Mautic, and schedules the outreach sequence.
5. Workflow 4 — Video Content Pipeline (Deploy Week 3–4): Sales team inputs content ideas into Google Sheets. n8n triggers the AI-Automated Shorts Generator workflow, monitors completion, and posts the finished video to Discord for review.
6. Workflow 5 — LinkedIn Outreach Cadence (Deploy Month 2): n8n manages the LinkedIn outreach schedule, enforcing daily rate limits, monitoring acceptance rates, and pausing automation if the acceptance rate drops below 20%.

Docker Compose Deployment

- Deploy n8n via Docker Compose with PostgreSQL as the database backend (not SQLite —

- PostgreSQL is required for production reliability)
- Use a reverse proxy (Nginx or Caddy) with SSL termination in front of n8n
- Set up automated PostgreSQL backups to S3 or equivalent — workflow data is critical
- Configure n8n's built-in credential encryption for all API keys stored in the platform
- Estimated infrastructure: \$20–40/month VPS + \$5–10/month managed PostgreSQL (or self-host PostgreSQL on the same VPS)

SECTION 7: THE FIRSTKNOCK POWER STACK — FINAL RECOMMENDATIONS

Master Comparison Table: All Tools Evaluated

Table 13 — FirstKnock Master Tool Comparison Matrix

Tool	Layer	Stars	Language	License	Integration Ease	FK Fit	Verdict
VROOM	Route Opt.	1,700	C++	Open Source	9/10	9/10	SHIP IT
Valhalla	Route Opt.	5,600	C++	MIT	8/10	10/10	SHIP IT
Nextroute	Route Opt.	23	Go	Source-Available	5/10	4/10	PASS
ColdFlow	Cold Email	7	TypeScript	GPL-3.0	3/10	2/10	DO NOT USE
CybrOutreach	Cold Email	0	Python	MIT	8/10	7/10	USE AS MICROSERVICE
cold-cli	Cold Email	0	Go	MIT	6/10	5/10	EARLY TESTING ONLY
OutreachStudio	Cold Email	38	N/A	BSL	1/10	1/10	DO NOT USE — VAPORWARE
AI-Auto-Short-Video-Gen	Video Gen	12	Python	None	6/10	6/10	REFERENCE ONLY
TikTokAIVideoGenerator	Video Gen	54	Python	CC0-1.0	7/10	8/10	SHIP IT

short-video-maker	Video Gen	24	Python	MIT	7/10	7/10	SHIP IT
tiktok-agent	Video Gen	13	Python/JS	MIT	7/10	7/10	SHIP IT (repurposing)
OpenOutreach	LinkedIn	~1,400	Python	Open Source	6/10	4/10	SECONDARY ACCOUNT ONLY
n8n (platform)	Automation	160K+	TypeScript	Fair-Code	9/10	10/10	SHIP IT — FOUNDATION
n8n Shorts	Video/Auto	N/A	n8n	N/A	8/10	9/10	SHIP IT
n8n B2B Lead Template	Email/Auto	N/A	n8n	N/A	8/10	8/10	SHIP IT

Top 5 Highest-Priority Integrations

1. Valhalla + VROOM (Route Optimization Stack) — Priority: CRITICAL
2. This is FirstKnock's core product differentiator. Every day a rep walks an unoptimized route is a day of lost revenue. Deploy this first, before anything else.
3. n8n (Automation Foundation) — Priority: CRITICAL
4. Without n8n, every other integration requires custom code. Deploy n8n Cloud on Day 1 and build all other integrations on top of it.
5. Mautic + Postal (Email Infrastructure) — Priority: HIGH
6. B2C homeowner nurture sequences are a key retention and trust-building mechanism. The post-door-knock email is the first touchpoint after a rep interaction — it must be automated and reliable.
7. TikTokAIVideoGenerator+ n8n Shorts Template (Content Pipeline) — Priority: HIGH
8. Content is the top-of-funnel for both rep recruitment and brand awareness. An automated content pipeline that produces 5 videos/week with zero manual effort is a significant competitive advantage.
9. CybrOutreach Microservice (B2B AI Personalization) — Priority: MEDIUM
10. Once the B2C nurture stack is running, the B2B outreach layer unlocks the dealer partnership channel. CybrOutreach's PAS framework + Anthropic API produces significantly higher response rates than generic cold emails.

Implementation Sequence

Table 14 — FirstKnock Implementation Sequence

Phase	Timeline	Deliverables	Tools Deployed
Phase 1	Week 1–2	Core infrastructure live. Reps receiving optimized routes. Door knock rightarrow email trigger working.	Valhalla (Docker), VROOM (Docker), n8n Cloud, Mautic, Postal, n8n B2B Lead Template
Phase 2	Week 3–4	Content pipeline live. B2B outreach running. CybrOutreach microservice deployed.	TikTokAIVideoGenerator , n8n Shorts Template, CybrOutreach FastAPI microservice, cold-cli (founder testing)
Phase 3	Month 2	LinkedIn automation on secondary account. n8n migrated to self-hosted. Full stack operational.	OpenOutreach (secondary account), n8n self-hosted Docker, tiktok-agent (repurposing pipeline), PostgreSQL backup automation

Total Cost Estimate

Table 15 — FirstKnock Monthly Infrastructure Cost Estimate

Component	Self-Hosted Monthly Cost	Cloud/SaaS Monthly Cost	Recommended Path
Valhalla + VROOM (VPS)	\$20–40	N/A (no cloud option)	Self-hosted from Day 1
n8n	\$5–20 (VPS add-on)	\$20–50 (n8n Cloud)	Cloud for Month 1, self-hosted Month 2+
Mautic	\$0 (same VPS)	\$0–50 (managed hosting)	Self-hosted on existing VPS
Postal	\$0 (same VPS)	N/A (self-hosted only)	Self-hosted from Day 1
CybrOutreach Microservice	\$0 (same VPS)	N/A (custom build)	Self-hosted from Day 1
AI APIs (OpenAI/Anthropic)	\$20–100 (usage-based)	\$20–100 (usage-based)	Pay-as-you-go
ElevenLabs (video voiceovers)	\$5–22/month	\$5–22/month	Starter plan

TOTAL ESTIMATE	\$50–180/month	\$65–242/month	Self-hosted stack is 25–30% cheaper at scale
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Non-Negotiable Stack vs. Nice-to-Have Stack

Table 16 — Non-Negotiable vs. Nice-to-Have Stack

Category	Tool	Justification
NON-NEGOTIABLE	Valhalla	Pedestrian routing intelligence. No substitute for OSM-aware walking routes.
NON-NEGOTIABLE	VROOM	VRP solver. Core product feature. Handles priorities and time windows.
NON-NEGOTIABLE	n8n	Automation backbone. Without it, every integration requires custom code.
NON-NEGOTIABLE	Mautic	B2C nurture automation. Post-door-knock email sequences are a core product feature.
NON-NEGOTIABLE	Postal	Deliverability infrastructure. Without it, emails go to spam.
NICE TO HAVE	CybrOutreach	B2B AI personalization. High value but not blocking core product launch.
NICE TO HAVE	TikTokAIVideoGenerator	Content pipeline. Important for growth but not core to rep workflow.
NICE TO HAVE	tiktok-agent	Content repurposing. Adds value once there is existing content to repurpose.
NICE TO HAVE	OpenOutreach	LinkedIn automation. High risk — only deploy on secondary account after core stack is stable.
SKIP	Nextroute	Source-available license + commercial upsell. VROOM is better for this use case.

SKIP	OutreachStudio	Vaporware. BSL license. 9 commits. No code available.
SKIP	ColdFlow	GPL-3.0 + potentially inaccessible. Do not build on it.

SECTION 8: RISKS & LIMITATIONS — DON'T BUILD ON SAND

Every architecture has failure modes. The FirstKnock stack is no exception. This section documents the specific risks associated with each tool and layer, with concrete mitigation strategies. Read this before you write a single line of integration code.

Abandoned & Vaporware Repos: The Graveyard

Table 17 — Vaporware and License Risk Registry

Repo	Risk Type	Specific Risk	Mitigation
OutreachStudio	Vaporware / BSL License	9 commits, no public code, Business Source License prohibits commercial use without a commercial agreement	DO NOT INTEGRATE. Monitor for 6 months — if code goes public and license changes, reassess.
ColdFlow	Abandoned / GPL-3.0	Repository potentially private/deleted. GPL-3.0 requires all derivative works to be GPL — incompatible with a commercial SaaS product.	DO NOT INTEGRATE. GPL-3.0 is a legal landmine for commercial products. Use Mautic + Postal instead.
AI-Automated-Short-Video-Generator	No License	No license specified means all rights reserved by default. Using this code in a commercial product without explicit permission is copyright infringement.	Use as a reference implementation only. Do not copy code directly. Use TikTokAIVideoGenerator (CC0) instead.

LinkedIn Automation: The Account Destruction Risk

EXISTENTIAL RISK: A permanent LinkedIn ban on the founder's account eliminates the startup's

primary B2B networking channel. This is not a recoverable situation. Treat LinkedIn automation with the same caution you would treat a production database migration — test on a secondary account first, always have a rollback plan, and never run at full volume without a proven warm-up period.

- **Account Restriction Rate:** 23% of automation users face restrictions within 90 days. That is nearly 1 in 4.
- **Enforcement Surge:** Account restrictions increased 340% since January 2026. LinkedIn is actively cracking down.
- **Permanent Ban Triggers:** Exceeding 80 connection requests/day, sending identical messages to 10+ people in an hour, or receiving multiple 'I don't know this person' reports
- **Mitigation Protocol:** Use OpenOutreach only on a dedicated secondary account. Never on the founder's primary account. Implement the 4-week safety ramp from Section 5. Monitor acceptance rates daily.
- **Emergency Protocol:** If the primary account receives a restriction warning, immediately cease all automation, manually review recent activity, and submit a LinkedIn appeal within 24 hours.

Technical Debt: C++ Build Complexity

Both Valhalla and VROOM are C++20 codebases. This is a strength (performance) and a risk (build complexity). If you ever need to compile from source — for a custom patch, a platform that doesn't support Docker, or a dependency conflict — you need a developer who is comfortable with CMake, C++ toolchains, and dependency management. For a 2–3 person team, this is a non-trivial ask.

- **Mitigation:** Always use the official Docker images for Valhalla and VROOM. Never compile from source unless absolutely necessary.
- **Mitigation:** Pin Docker image versions in your docker-compose.yml. Do not use 'latest' tags in production — a breaking change in a new release can take down your routing layer.
- **Mitigation:** Subscribe to the Valhalla and VROOM GitHub release notifications. Review changelogs before updating.
- **Mitigation:** Maintain a tested rollback procedure — if a new Docker image breaks routing, you need to be able to revert to the previous version in under 5 minutes.

Technical Debt: Go Binary Management for cold-cli

cold-cli's single Go binary is an elegant deployment model, but it introduces a different kind of technical debt: the binary must be recompiled for each target platform (Linux x86_64, ARM, macOS), and the Gmail API dependency via gws means any changes to Google's OAuth flow can break the tool without warning. The 'Agent-first' design (built with Claude Code) is innovative but means the tool's development velocity is tied to a single developer's availability.

- **Mitigation:** Use cold-cli only for founder-led early outreach testing. Do not build production workflows that depend on it.

- Mitigation: Monitor the repo for Gmail API breakage — if Google changes OAuth scopes, the tool will stop working until the maintainer patches it.
- Mitigation: Transition to Mautic + Postal for all production sending before scaling beyond 50 emails/day.

License Risk Matrix

Table 18 — License Risk Matrix

License Type	Tools	Commercial Use	Modification	Distribution Risk	Verdict
MIT	Valhalla, CybrOutreach, cold-cli, short-video-maker,	Yes—unrestricted	Yes—unrestricted	None	SAFE — use freely
CC0-1.0 (Public Domain)	tiktok-agent TikTokAIVideo Generator	Yes—unrestricted	Yes—unrestricted	None	SAFE — best possible license
Open Source (Verso)	VROOM	Yes—check specific terms	Yes	Low	SAFE — verify specific terms
GPL-3.0	ColdFlow	Yes, but derivative works must be GPL	Yes, but must release source	HIGH — copyleft infection risk	AVOID for commercial SaaS
Business Source License (BSL)	OutreachStudio	No — requires commercial license	Restricted	HIGH — not open source	DO NOT USE
Source-Available (NOASSERTIO	Nextroute	Restricted — requires Nextmv Cloud	Restricted	HIGH — vendor lock-in	DO NOT USE
No License Specified	AI-Auto-Short-Video-Gen	No — all rights reserved by default	No	HIGH — copyright infringement risk	REFERENCE ONLY — do not copy code

Dependency & Ecosystem Risks

- Valhalla OSM Tile Data: OpenStreetMap data quality varies by region. In rural or newly developed areas, pedestrian path data may be incomplete. Validate tile quality for each new

market before deploying to reps.

- ElevenLabs API (Video Pipeline): ElevenLabs is a commercial SaaS with usage-based pricing. A viral content campaign could generate unexpected API costs. Set hard spending limits in the ElevenLabs dashboard.
- Groq API (Video Generation): Groq's free tier has rate limits that will be hit quickly in a high-volume content pipeline. Budget for paid Groq API access or implement a fallback to OpenAI.
- Mautic PHP Stack: Mautic is PHP-based, which adds a PHP runtime dependency to the stack. Ensure your VPS has PHP 8.1+ and the required extensions. Consider using the official Mautic Docker image to avoid PHP version conflicts.
- n8n Fair-Code License: n8n uses a 'Fair-Code' license (not fully open source). The self-hosted Community Edition is free, but certain enterprise features require a paid license. Verify that all features you need are available in the Community Edition before committing to self-hosting.

Summary: Risk Mitigation Priorities

1. IMMEDIATE: Remove OutreachStudio and ColdFlow from all integration plans. They are legal and technical liabilities.
2. IMMEDIATE: Establish a dedicated secondary LinkedIn account for all automation testing. Never run OpenOutreach on the founder's primary account.
3. WEEK 1: Pin all Docker image versions in docker-compose.yml. Set up automated backup for n8n PostgreSQL database.
4. WEEK 2: Configure spending limits on all AI APIs (OpenAI, Anthropic, ElevenLabs, Groq) to prevent unexpected cost spikes.
5. MONTH 1: Validate Valhalla OSM tile quality for each new market before deploying optimized routes to reps in that market.
6. ONGOING: Monitor GitHub release notifications for Valhalla, VROOM, and n8n. Review changelogs before any version updates in production.

This audit was prepared for internal use by the FirstKnock engineering and product team. All GitHub statistics are verified as of April 2026. Tool assessments reflect the state of each repository at the time of audit. The open-source ecosystem moves fast — reassess any tool rated below 5/10 on maturity before integrating, as project status may have changed.

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